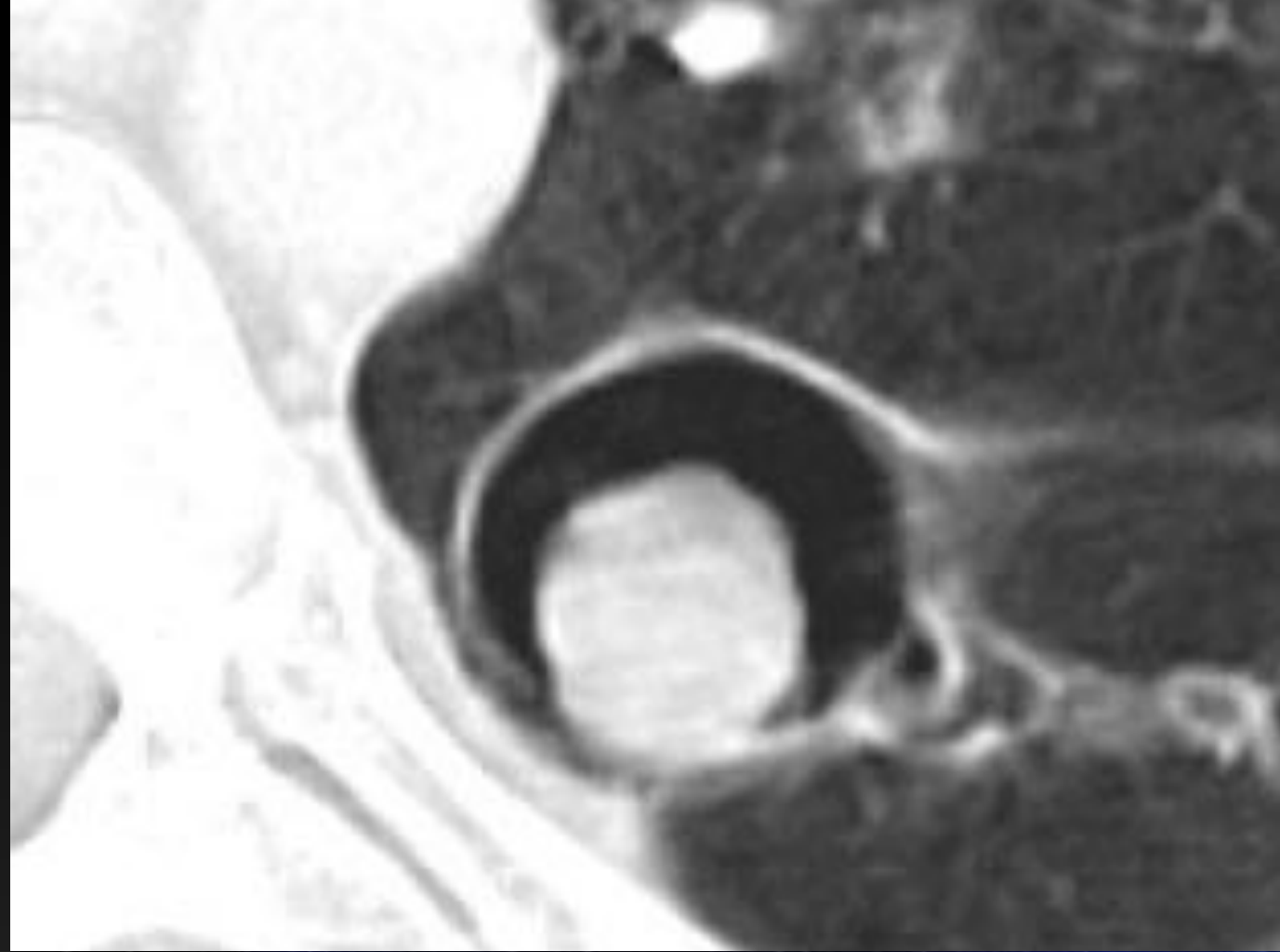


Achados Radiológicos do Espectro da Aspergilose

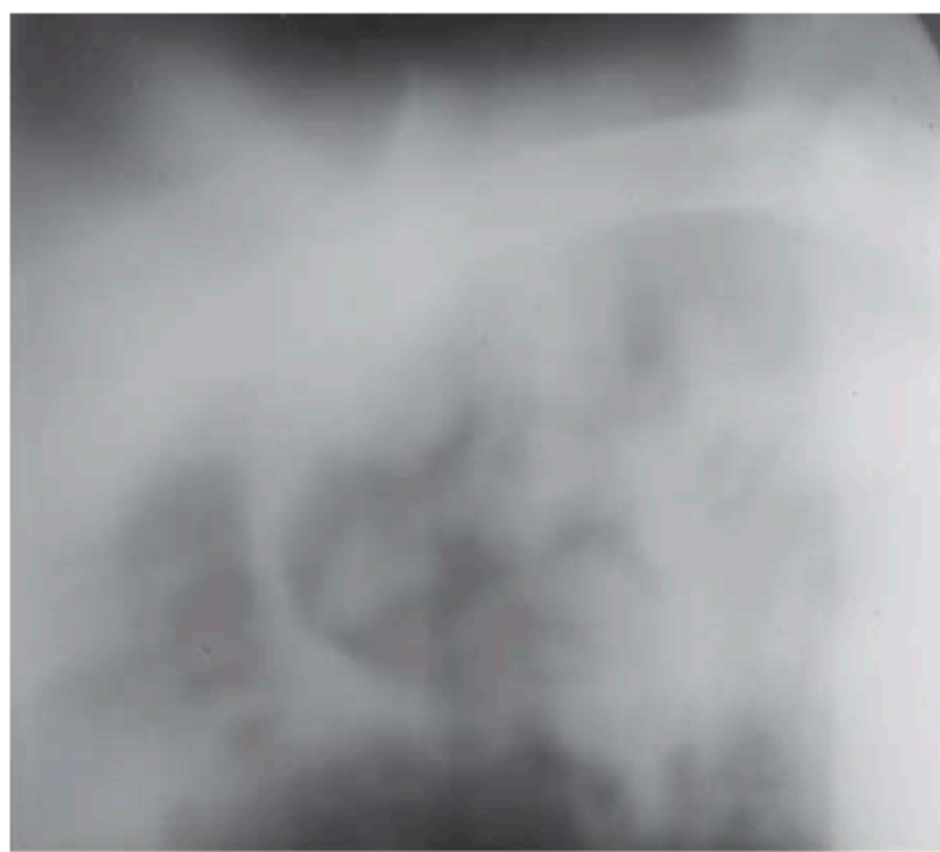
Aspergilose Saprofítica (Aspergiloma)

crescente aéreo



Conglomerados de hifas misturado com muco e restos celulares em uma cavidade preexistente

Aspergilomas estão associados a espessamento da parede da cavidade e podem ser o seu primeiro sinal radiográfico



a.



b.

Figure 2. Aspergilloma in a 54-year-old man with a history of tuberculosis. (a) Linear tomogram (magnified view) shows multiple fungus balls within a cavity in the right upper lobe. (b) Photograph of the corresponding gross surgical specimen demonstrates multiple irregular fungus balls virtually filling the pulmonary cavity.

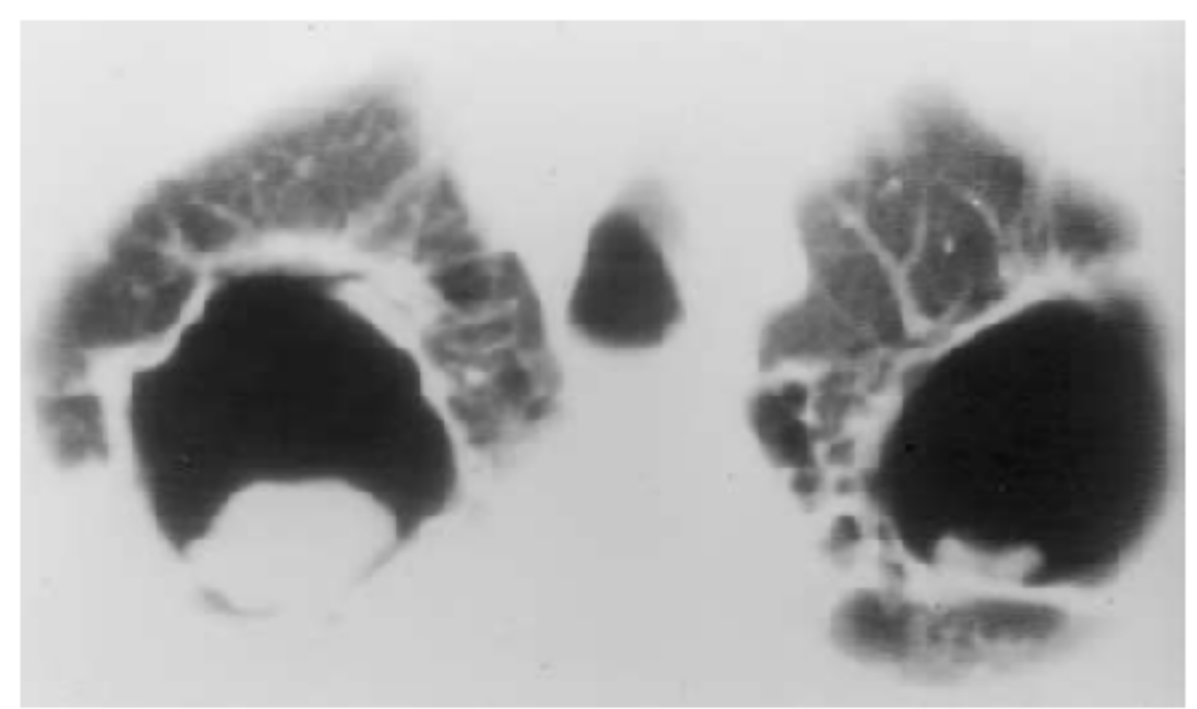


Figure 3. Bilateral aspergillomas in a 71-year-old man with residual tuberculosis. Chest computed tomographic (CT) scan (lung window) shows large cavities bilaterally in the upper lobes containing fungus balls of different sizes.

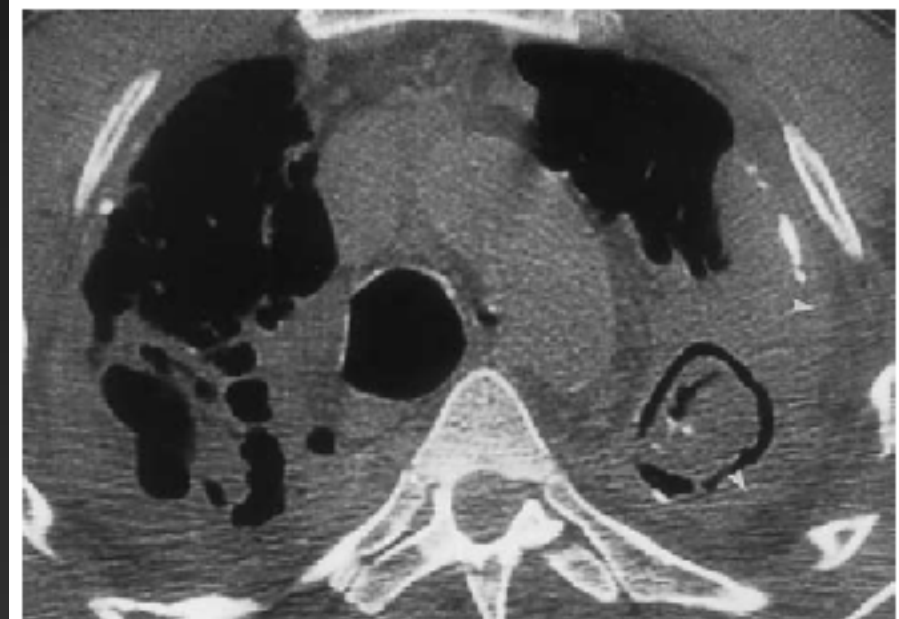
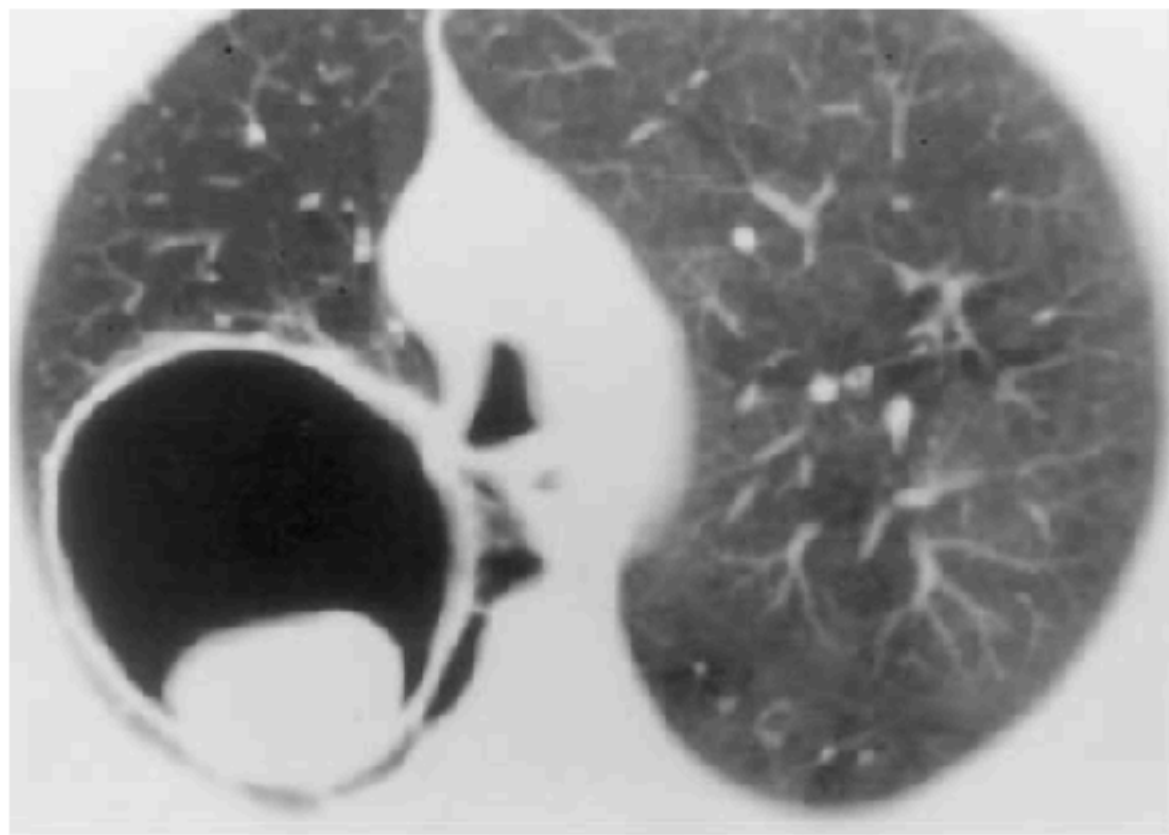
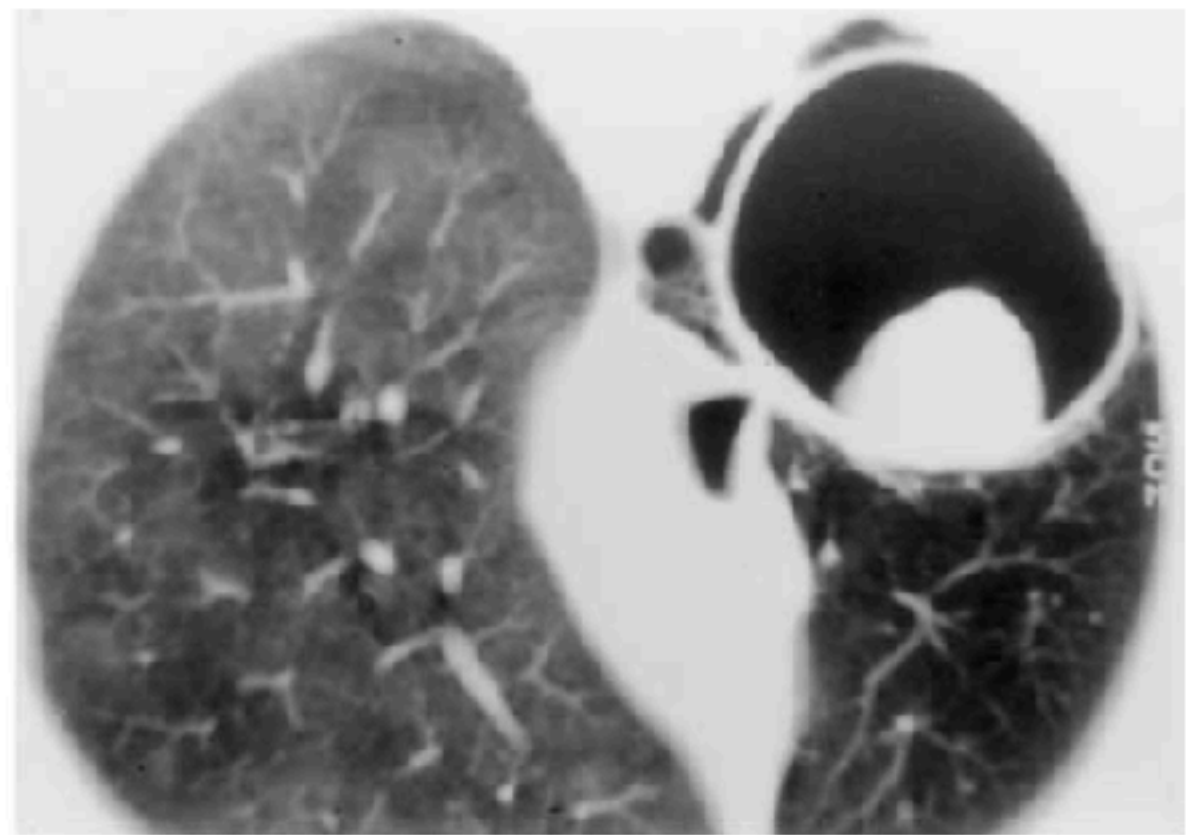


Figure 4. Aspergilloma with the air crescent sign in a 67-year-old woman with residual tuberculosis. Thin-section CT scan (mediastinal window) shows extensive scarring in the upper lobes. A large aspergilloma is seen in the left upper lobe, with a characteristic air crescent between the aspergilloma and the cavity wall. Note the marked pleural thickening surrounding the cavity containing the aspergilloma (arrowheads). Punctate bilateral pleural calcifications are also visible.



a.



b.

Figure 5. Mobile aspergilloma within a pulmonary cystic cavity in a 43-year-old man. Chest CT scans obtained with the patient supine (**a**) and prone (**b**) show a change in the position of the aspergilloma. *A fumigatus* was discovered at bronchoscopy. (Courtesy of Josep M. Mata, MD, Unidad Diagnóstica de Alta Tecnología, Sabadell, Spain.)

**Reação de
Hipersensibilidade
(Aspergilose
Broncopulmonar Alérgica)**

dedos em luva



Plugues de muco contendo aspergillus e eosinófilos, que causam dilatação de brônquios segmentares e subsegmentares

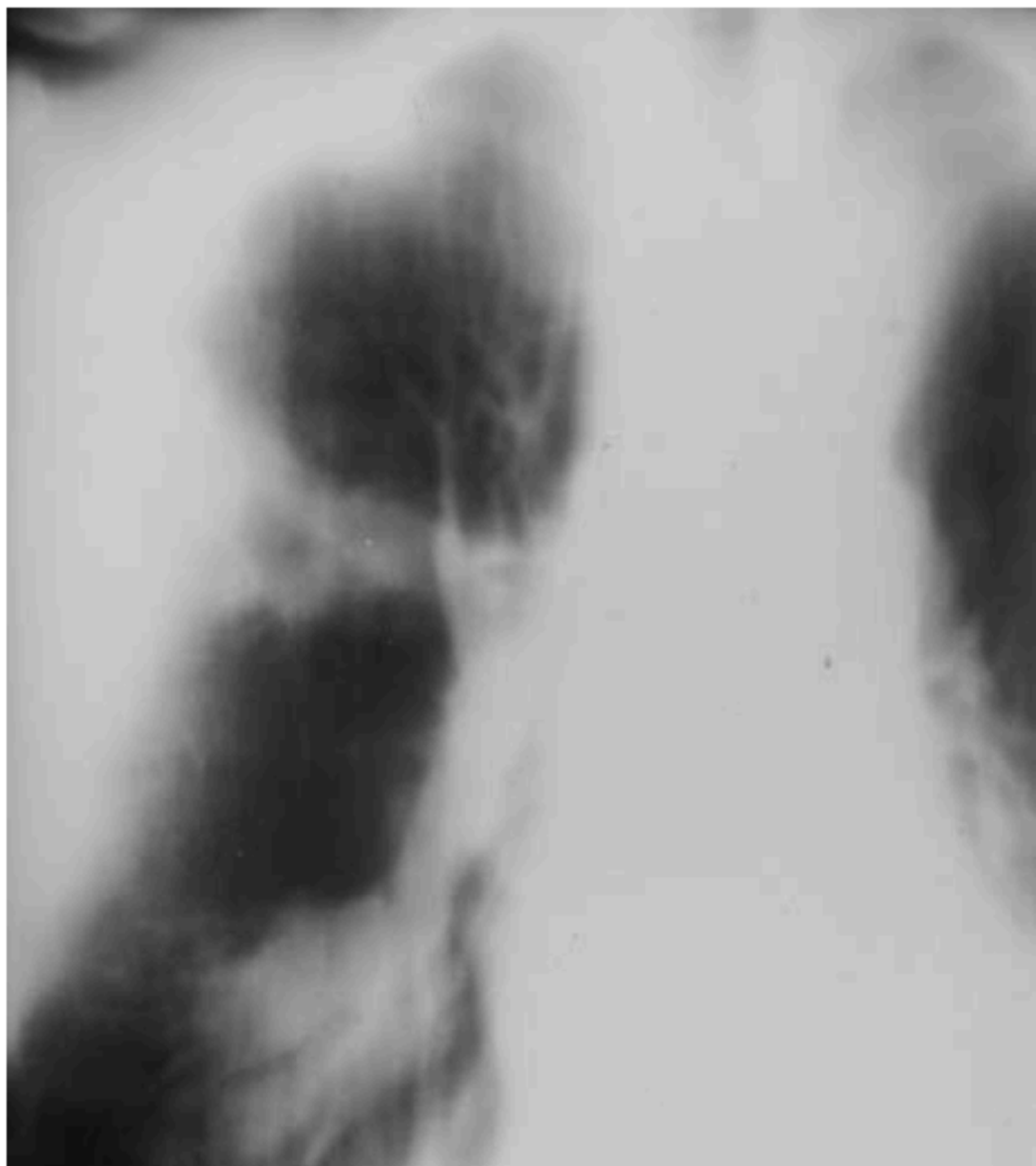


Figure 7. Allergic bronchopulmonary aspergillosis in a 32-year-old asthmatic patient. Linear tomogram (magnified view) of the right lung reveals multiple tubular shadows extending in a distribution corresponding to the bronchovascular bundles. These Y-shaped shadows are caused by inspissated mucus within markedly dilated bronchi. (Courtesy of Joaquín Fernández-Cruz, MD, Hospital Virgen del Rocío, Seville, Spain.)

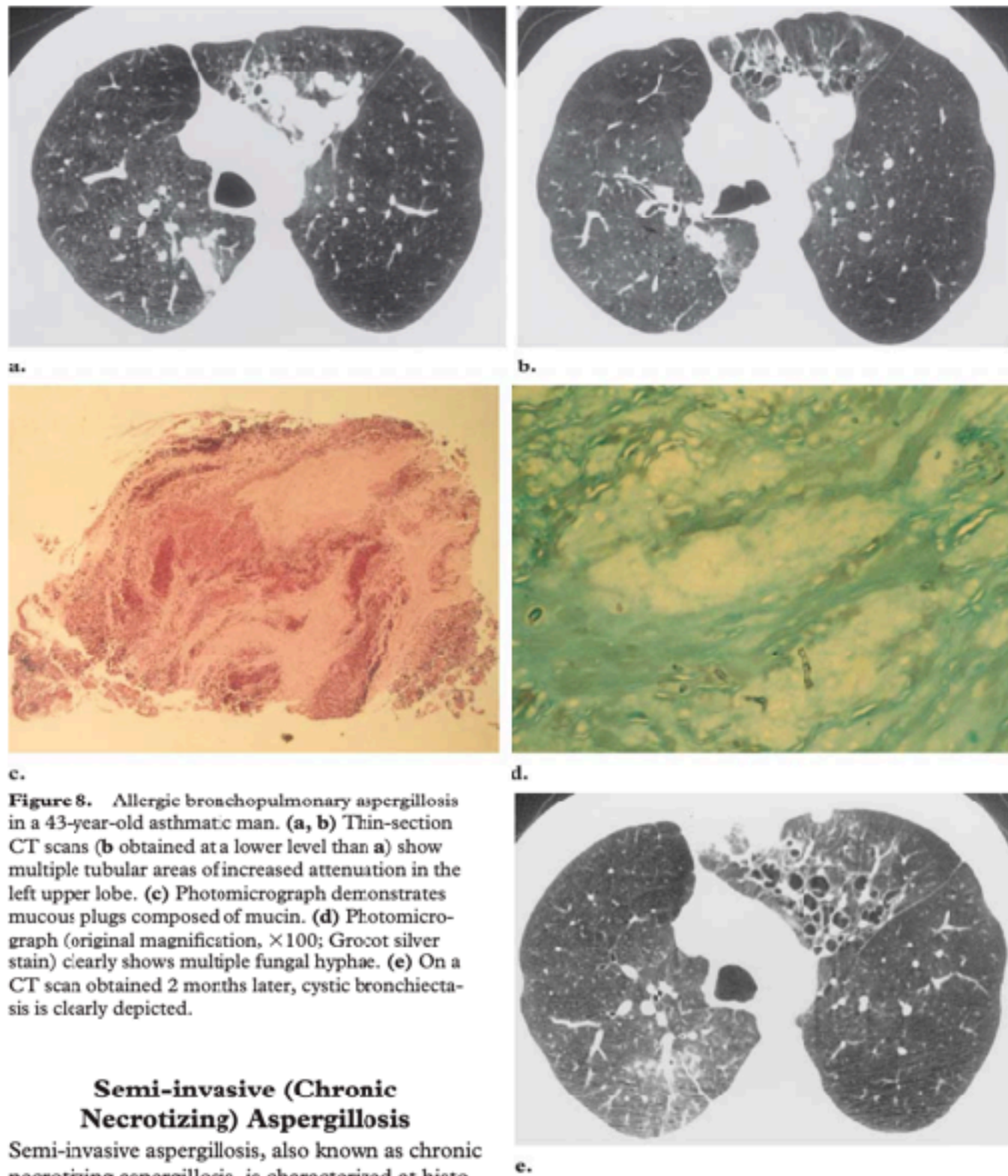


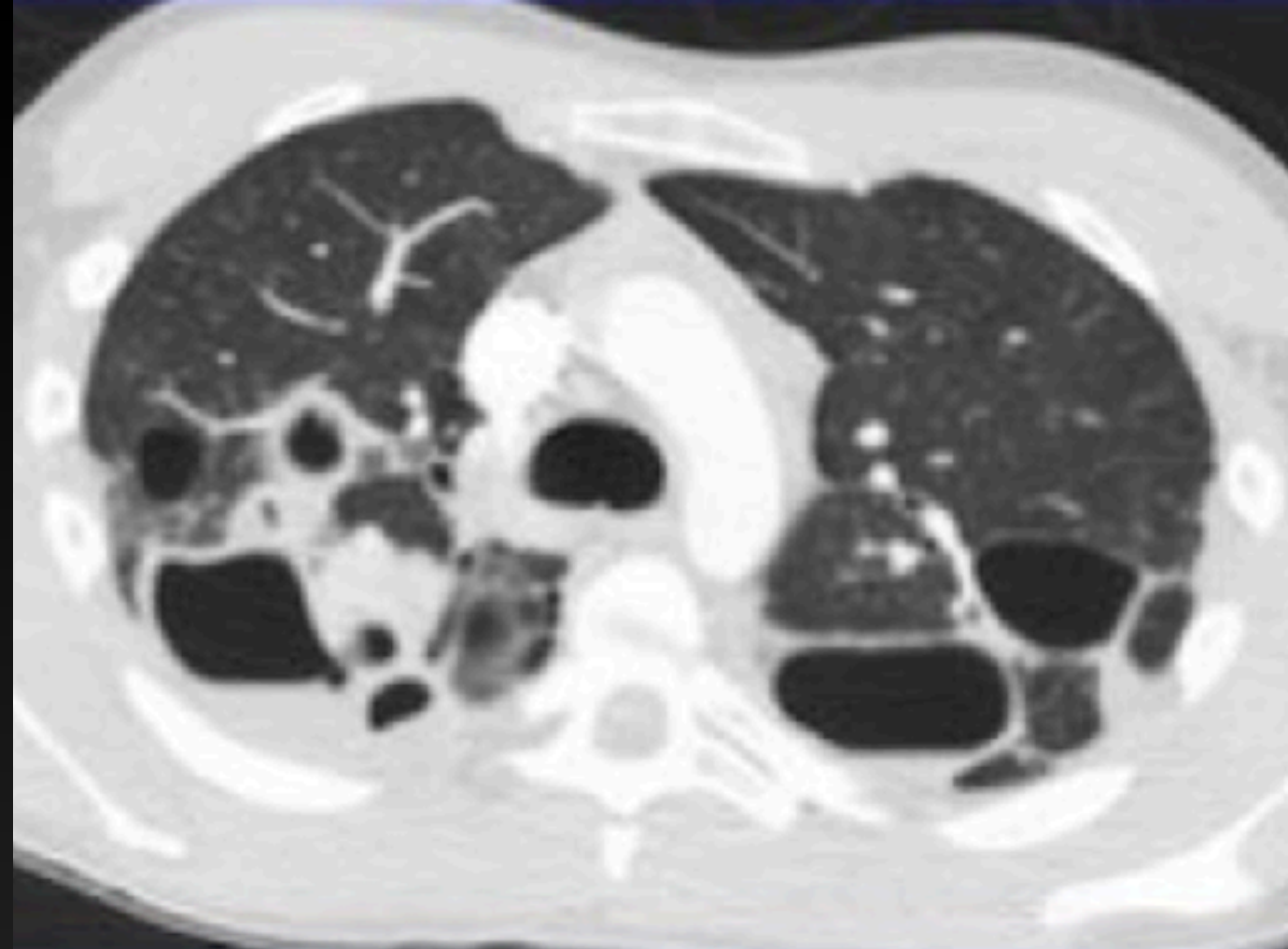
Figure 8. Allergic bronchopulmonary aspergillosis in a 43-year-old asthmatic man. (a, b) Thin-section CT scans (b obtained at a lower level than a) show multiple tubular areas of increased attenuation in the left upper lobe. (c) Photomicrograph demonstrates mucous plugs composed of mucin. (d) Photomicrograph (original magnification, $\times 100$; Grocott silver stain) clearly shows multiple fungal hyphae. (e) On a CT scan obtained 2 months later, cystic bronchiectasis is clearly depicted.

Semi-invasive (Chronic Necrotizing) Aspergillosis

Semi-invasive aspergillosis, also known as chronic necrotizing aspergillosis, is characterized at histo-

**Aspergilose Semi-
invasiva (Necrotizante
Crônica)**

*consolidações
escavadas*

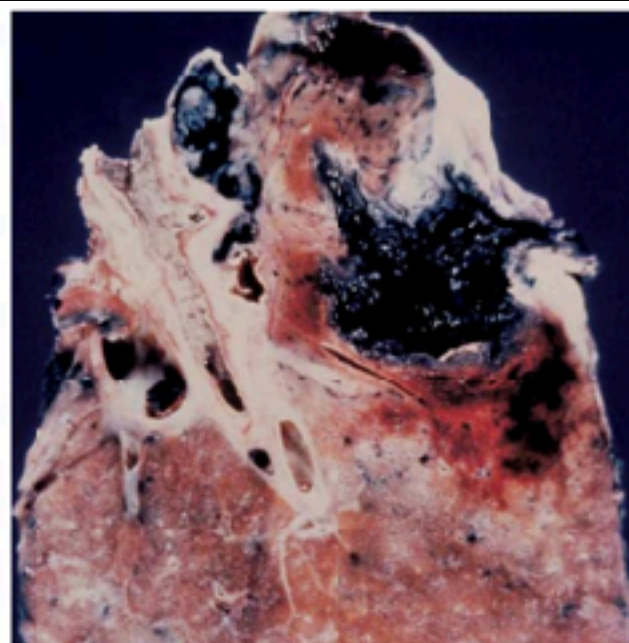


**Necrose tecidual e reação granulomatosa
similar à reativação de tuberculose**

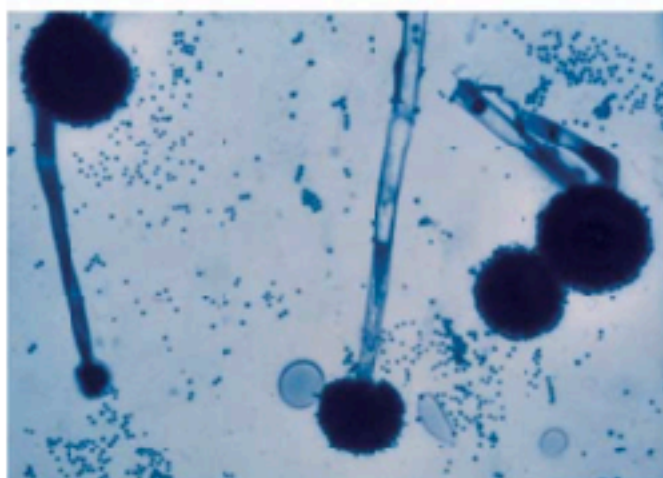
Predisposta por condições debilitantes



a.

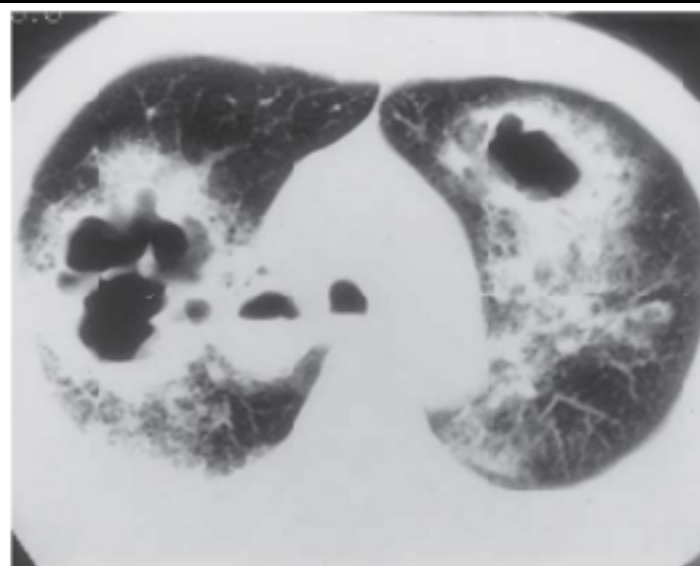


b.

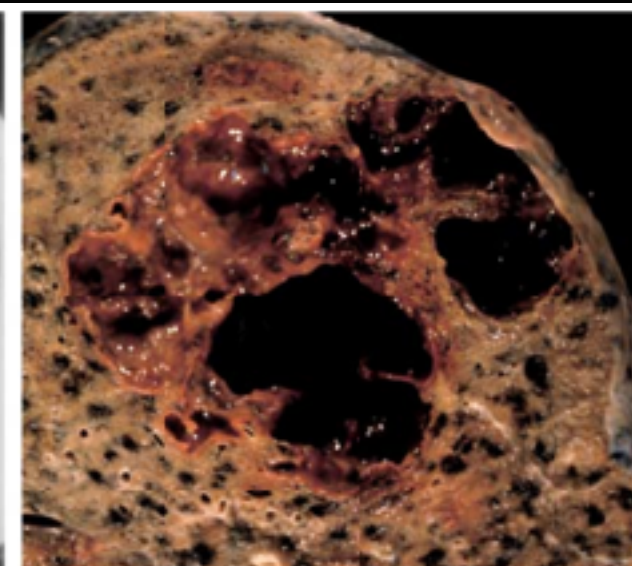


c.

Figure 10. Semi-invasive aspergilloma in a 54-year-old alcoholic man with chronic bronchitis and recurrent episodes of hemoptysis. (a) Linear tomogram (magnified view) shows a large cavity containing multiple fungus balls in the left upper lobe with associated adjacent pleural thickening. (b) Photograph of an autopsy specimen from the left upper lobe shows an irregular cavitory lesion with regular margins and a dark brown appearance caused by necrotic material and superimposed *Aspergillus* infection. (c) High-power photomicrograph shows massive *A. niger* with its characteristic black appearance.

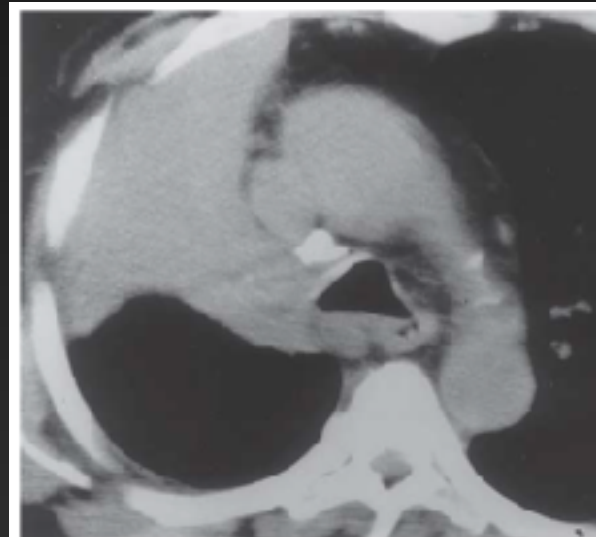


a.

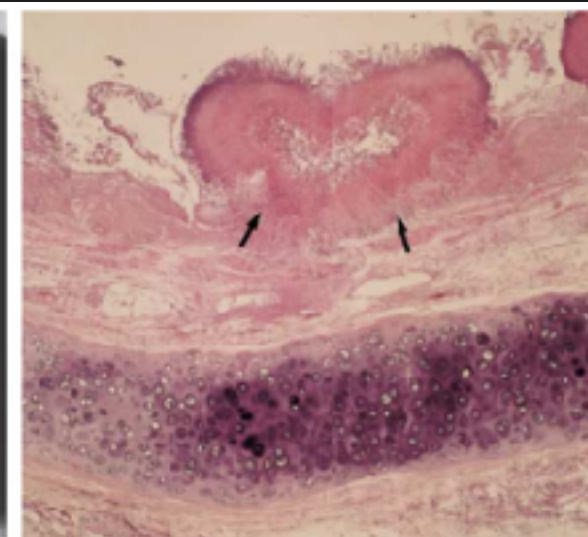


b.

Figure 9. Semi-invasive aspergillosis in a 68-year-old man with chronic bronchitis and recurrent episodes of mild hemoptysis. (a) Thin-section CT scan (lung window) shows bilateral rounded areas of consolidation with associated cavitation in both upper lobes. (b) Photograph of an autopsy specimen from the left upper lobe shows an irregular cavitory lesion with regular margins and a dark brown appearance caused by necrotic material and *Aspergillus* infection.

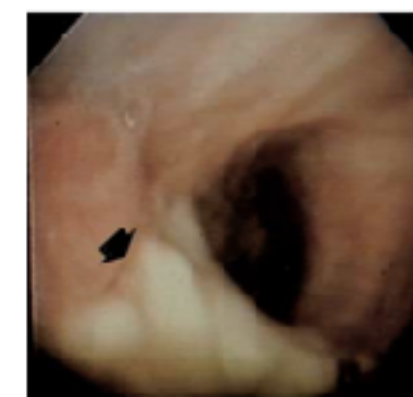


a.



b.

Figure 11. Necrotizing bronchial aspergillosis in a 54-year-old man who presented with cough and sputum production. (a) Chest CT scan (mediastinal window) obtained at the level of the carina shows a thickened, narrowed right main bronchus with associated right upper lobe collapse. (b) Bronchoscopic image shows elevated, whitish mucosal lesions in the right main bronchus (arrow). (c) High-power photomicrograph (original magnification, $\times 400$; hematoxylin-eosin stain) of a biopsy specimen from the right upper lobe reveals massive *Aspergillus* hyphae invading the bronchial epithelium (arrows).

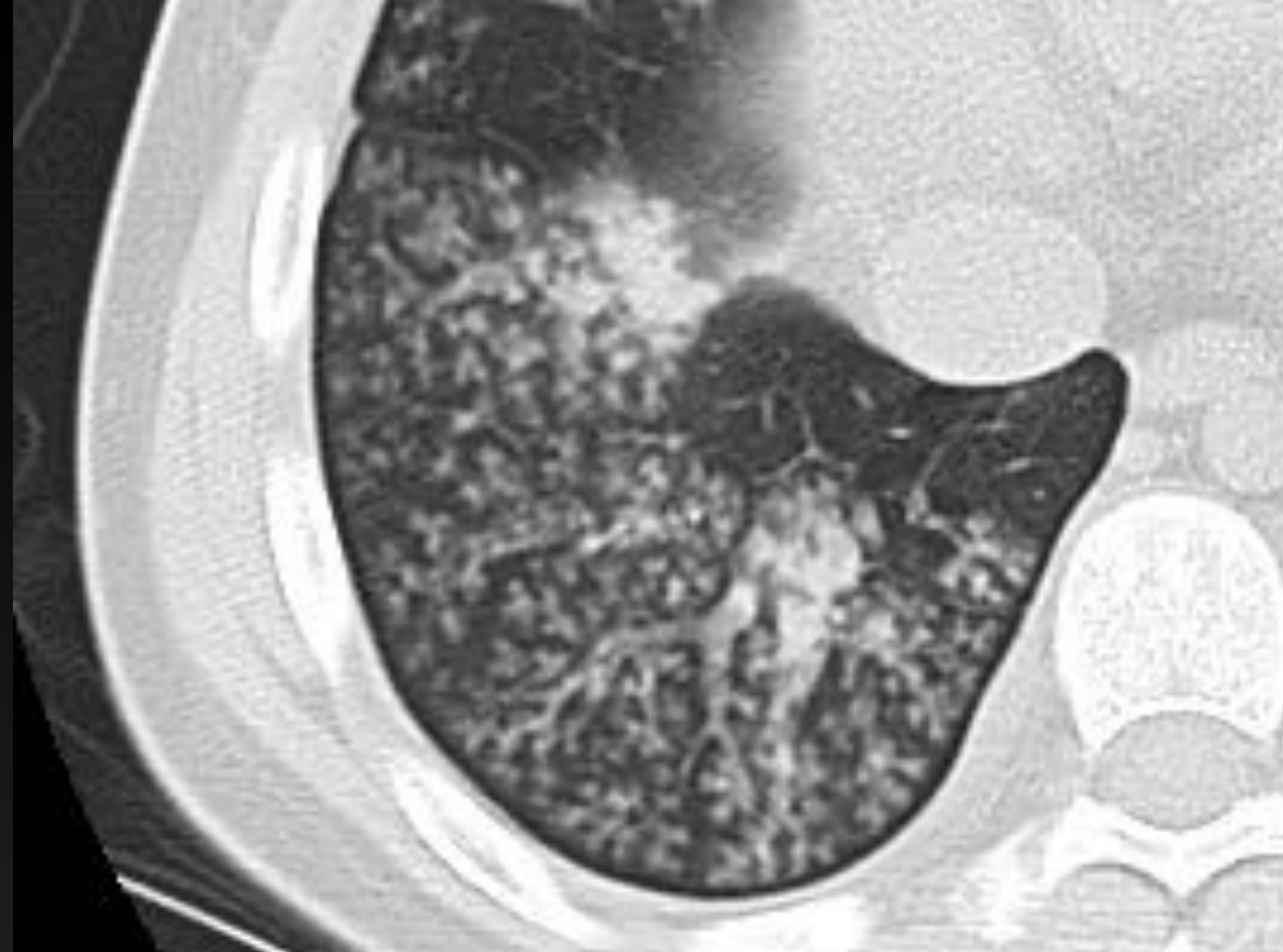


c.

Aspergillus necrotizing bronchitis may be seen as an endobronchial mass, obstructive monitis or collapse, or a hilar mass. Only a few reports have described CT findings in *As-*

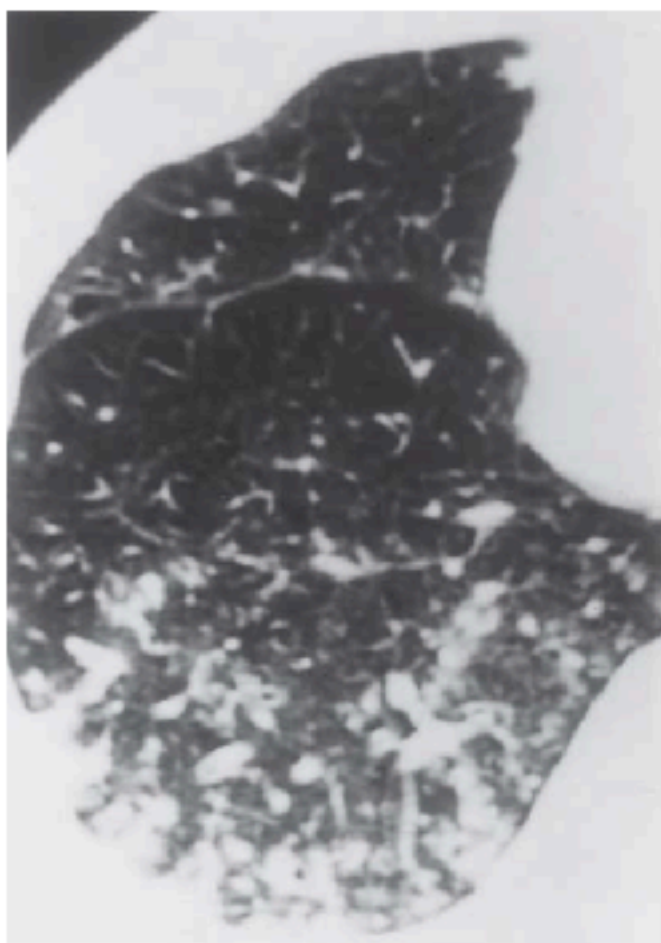
Aspergilose Invasiva das Vias Aéreas

árvore em brotamento



**Penetração de Aspergillus na via aérea,
até a membrana basal**

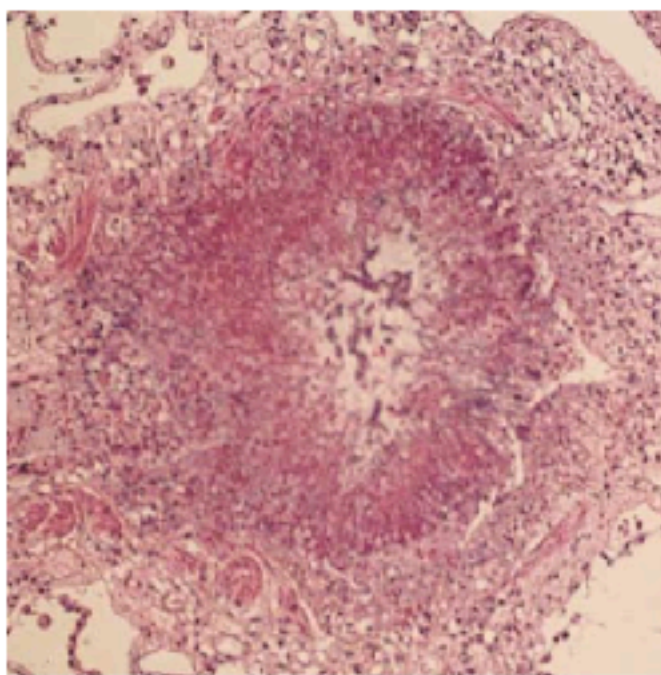
Mais comum em imunocomprometidos



a.



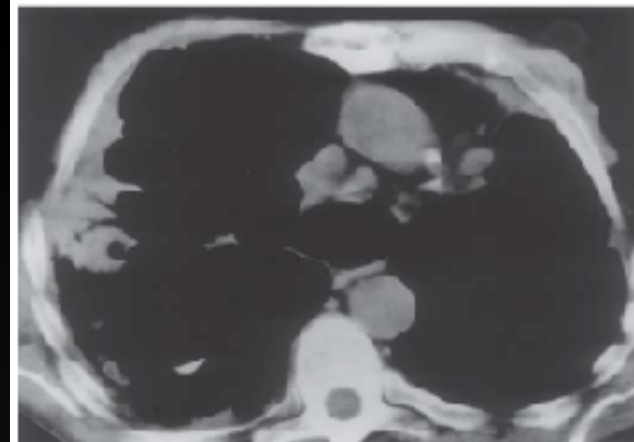
b.



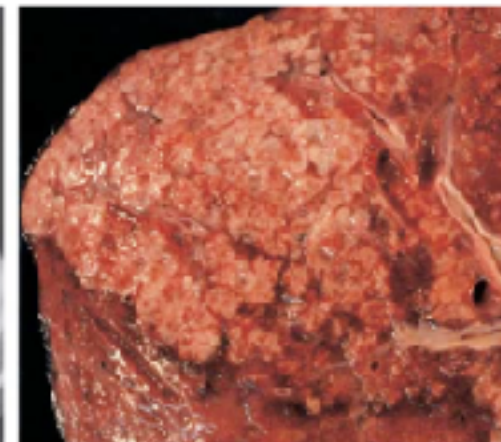
c.

Figure 12. Invasive bronchiolar aspergillosis in a patient who had undergone bone marrow transplantation. (a) Thin-section CT scan (lung window) shows peripheral branching structures associated with focal areas of consolidation in the right lower lobe. (b) Photograph of the corresponding autopsy specimen shows multiple yellowish acinar nodules. (c) High-power photomicrograph of a lung biopsy specimen demonstrates complete destruction of the bronchiolar wall by *Aspergillus* infection.

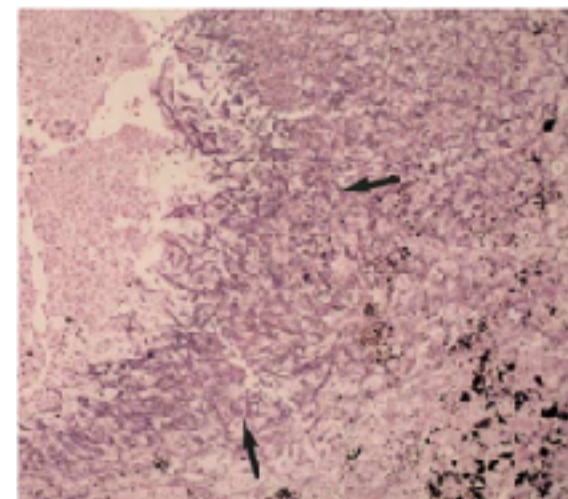
Centrilobular nodular areas of intensity similar to those seen in *Aspergillus* have been described in a number of cases, including endobronchial spread of tuberculosis. *Mycobacterium avium*



a.



b.

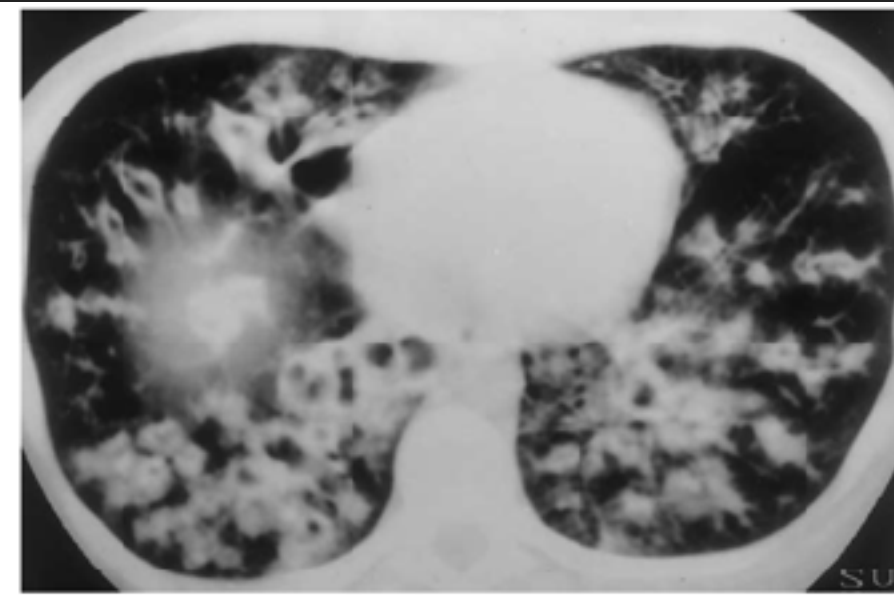


c.

Figure 13. Bronchopneumonia aspergillosis. (a) Conventional CT scan through the upper lungs shows a segmental area of consolidation in the right upper lobe with visible air bronchogram. (b) Photograph of the corresponding autopsy specimen shows segmental consolidation. (c) High-power photomicrograph of a small area of consolidation shows tissue necrosis. Scattered *Aspergillus* organisms can be identified in the necrotic tissue (arrows).

Obstructing bronchopulmonary aspergillosis is a noninvasive form of aspergillosis characterized by the massive intraluminal overgrowth of *Aspergillus* species, usually *A. fumigatus*, in patients with AIDS (18). Affected patients exhibit cough, fever, and new onset of asthma. Patients may cough up fungal casts of the bronchi and present with severe hypoxemia. The characteristic CT

Figure 14. Obstructing bronchopulmonary aspergillosis in a 29-year-old man with AIDS. CT scan shows multiple rounded and tubular areas of increased attenuation in both lower lobes, findings that are consistent with mucus-filled airways. Bronchoscopy revealed that the lumen was packed with inflammatory material.



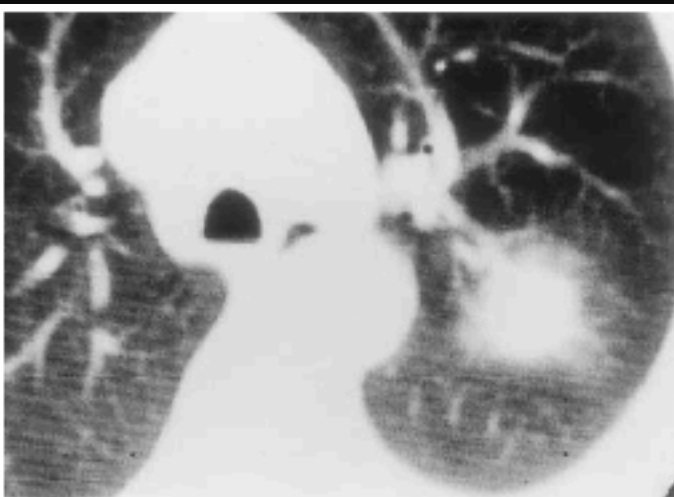
SU

Aspergilose Angioinvasiva

halo



**Invasão e oclusão de artérias
médias e pequenas pelas hifas**

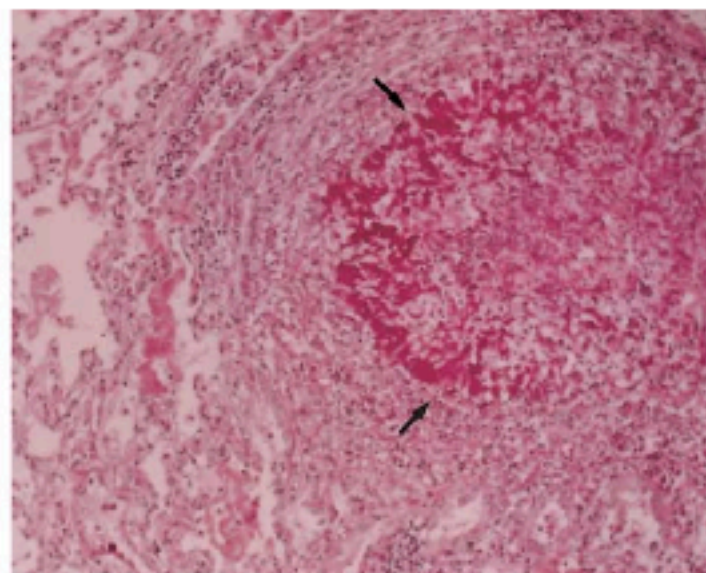


a.

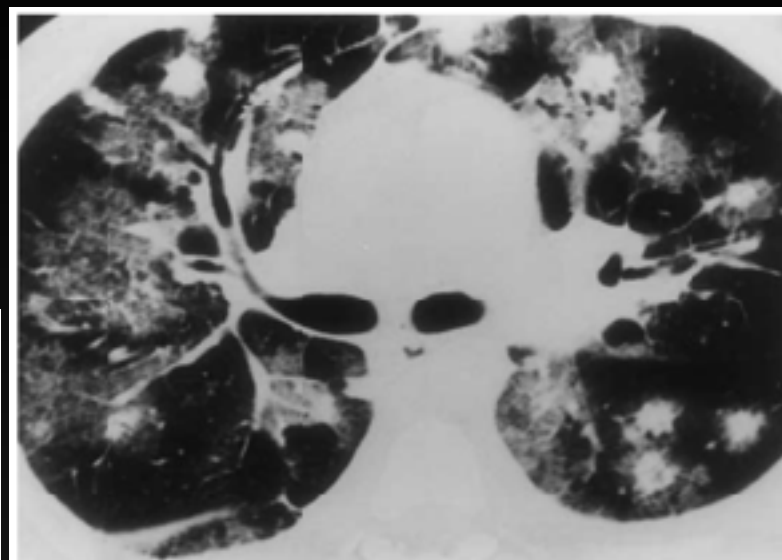
Figure 15. Angioinvasive aspergillosis in a 42-year-old man with acute myelogenous leukemia. (a) Chest CT scan (lung window) reveals a 2-cm nodular lesion with a wide halo of ground-glass attenuation representing adjacent hemorrhage. (b) Photograph of a cut section of the lung demonstrates a rounded tan nodule, a finding that is consistent with pulmonary infarction. (c) Low-power photomicrograph (original magnification, $\times 40$; hematoxylin-eosin stain) shows vascular invasion by *Aspergillus* species (arrows).



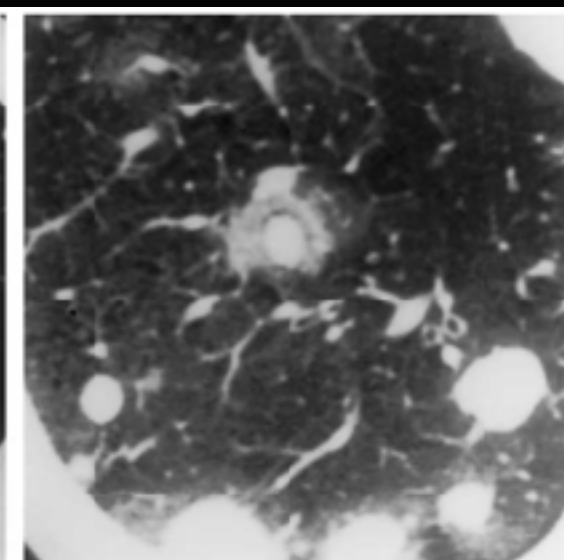
b.



c.

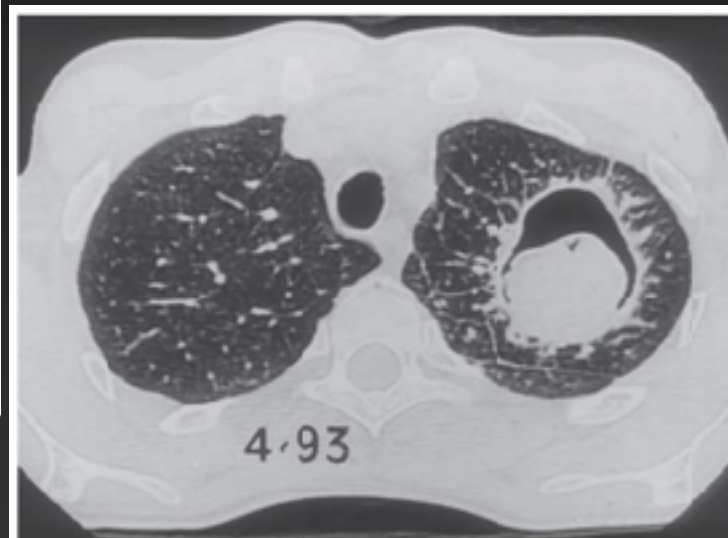


16.

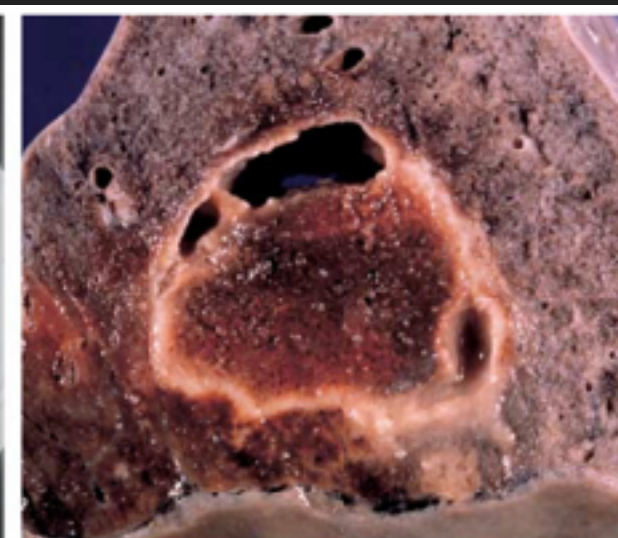


17.

Figures 16, 17. (16) Cutaneous Kaposi sarcoma in a 34-year-old homosexual man with AIDS. Thin-section CT scan shows multiple hemorrhagic pulmonary nodules with an associated halo sign. (17) Hemorrhagic metastases in a 62-year-old man. High-resolution CT scan demonstrates multiple lung nodules, some of them with surrounding halos of ground-glass attenuation representing hemorrhage.



a.



b.

Figure 18. Angioinvasive aspergillosis in a 54-year-old man. (a) Thin-section CT scan (lung window) shows a cavitated nodule with air crescent formation. (b) Photograph of the pathologic specimen shows a thick-walled cavity with corresponding air crescent formation.

Early diagnosis of invasive pulmonary aspergillosis in hematologic patients: an opportunity to improve the outcome

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Invasive pulmonary aspergillosis (IPA) is the leading invasive fungal disease in high-risk hematologic patients, including those with acute myeloid leukemia (AML) receiving chemotherapy for induction of remission, and allogeneic hematopoietic cell transplant (HCT) recipients.^{1,2} Studies conducted in the 1990s and early 2000s reported crude mortality

disease. An important contribution to the management of IPA was made by studies showing the importance of the halo sign as the earliest detectable sign of disease. Caillot *et al.* analyzed the diagnosis of IPA in two periods. In the first, high-resolution chest computed tomography (CT) was obtained at the discretion of clinicians, based on clinical signs of infection, whereas

Conceitos

1. Mortalidade diretamente relacionada à carga fúngica
2. Carga fúngica diretamente relacionada à forma da doença (na imagem também)
3. Achados inespecíficos podem ser a fase inicial da doença invasiva, em um contexto apropriado
4. O tratamento instituído em uma fase precoce diminui a mortalidade

Pathological changes in the lungs

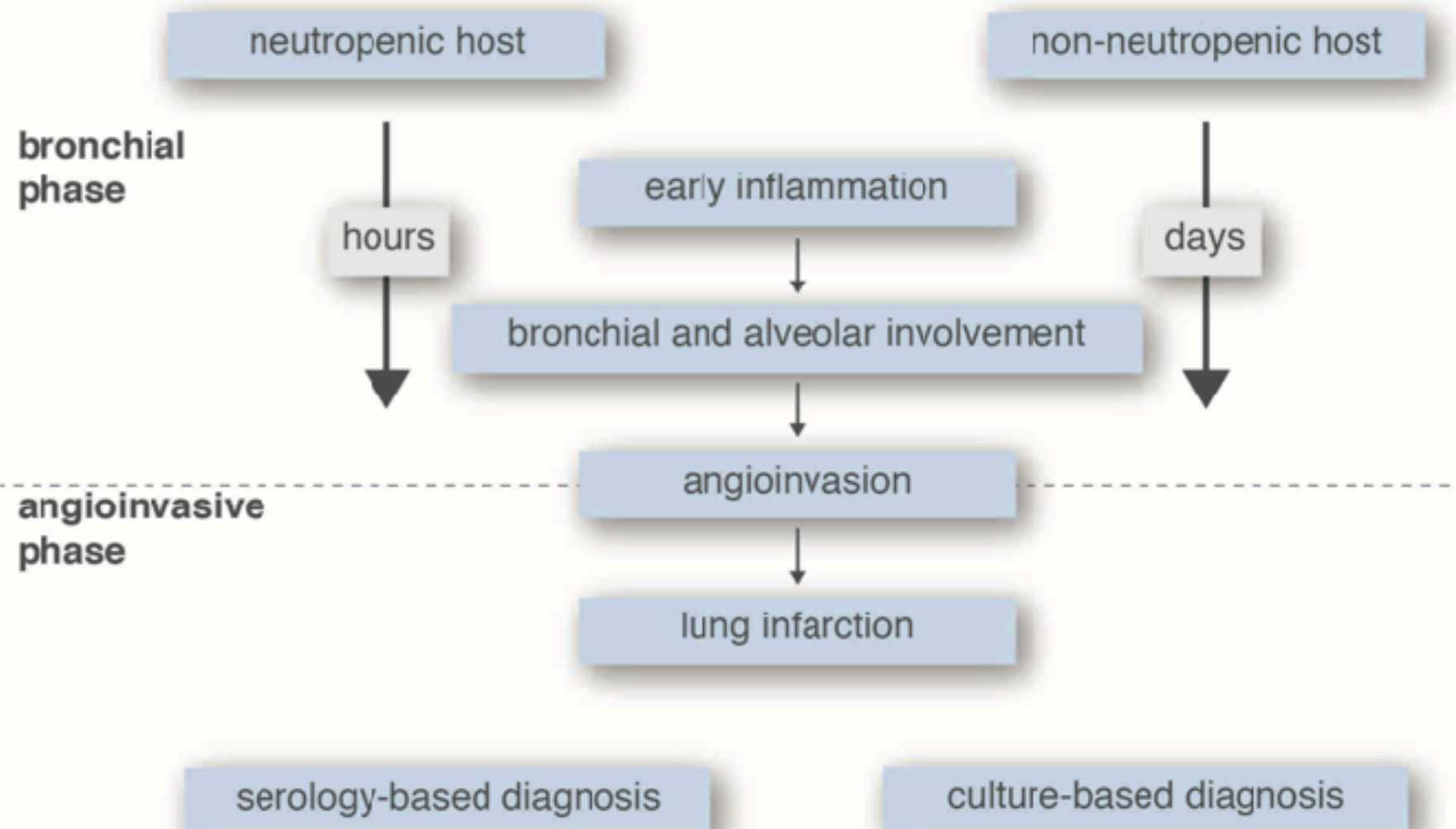
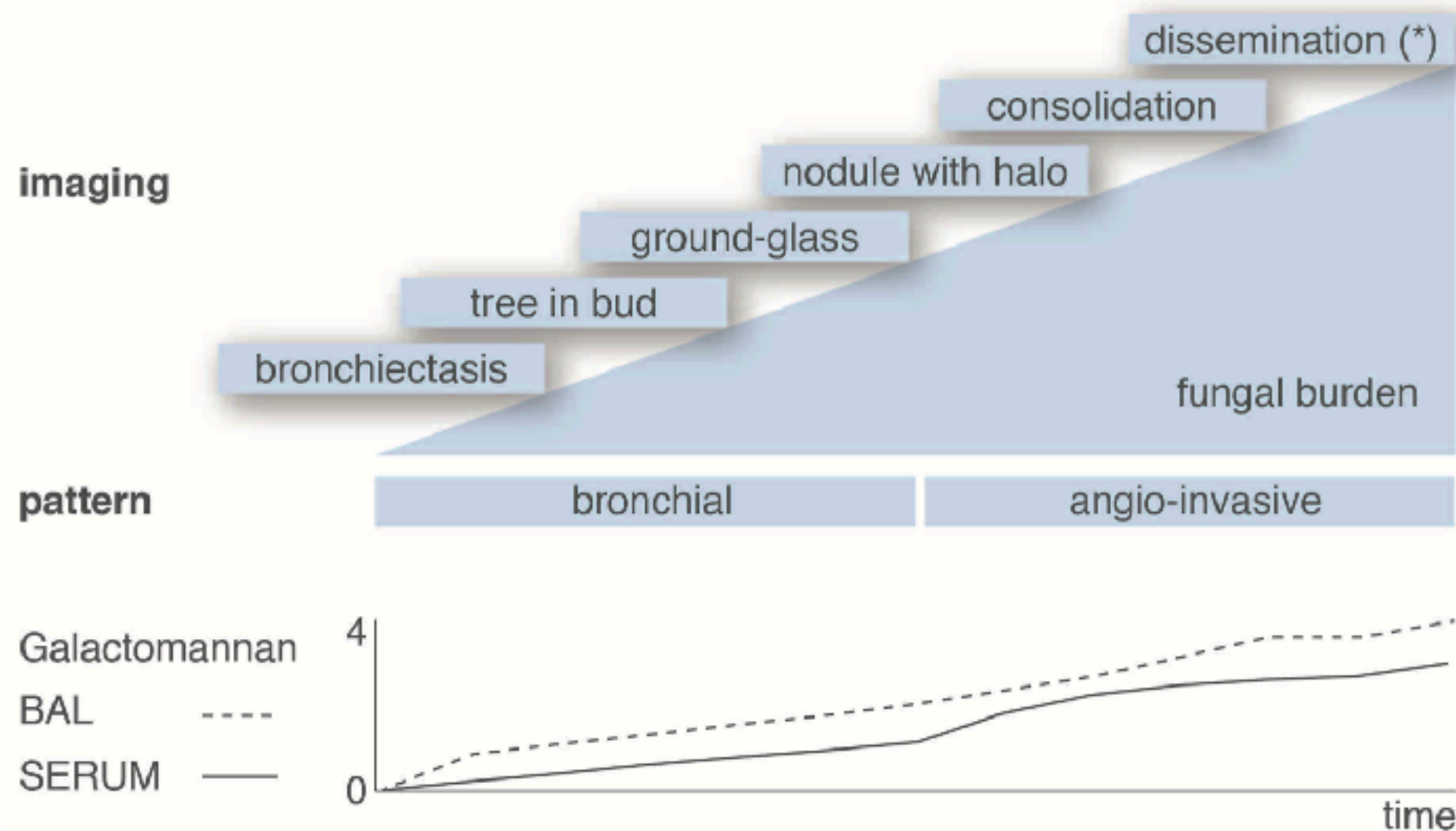


Figure 1. Evolution of invasive pulmonary aspergillosis in the bronchoalveolar and the angioinvasive phases in neutropenic and non-neutropenic patients.



* lesions evolve to air crescent and cavity when neutrophils recover
BAL: bronchoalveolar lavage.

Figure 2. The clinical evolution of invasive pulmonary aspergillosis. *Lesions evolve to air crescent and cavity when neutrophils recover. BAL: bronchoalveolar lavage.